

IG Gematria: Numerical Vector Operations on the Inscribing Grammar Catalog

May 30, 2026

Author: Lando $\otimes$ $\odot$ operator

0.1 Abstract

We treat the 2,858 valid entries of the Inscribing Grammar catalog as 12-dimensional integer vectors over the ordinal encoding $\{0, \dots, \text{card} - 1\}$ per primitive. This enables **true gematria** — the systematic investigation of additive equations $A + B = C$, multiplicative (Hadamard) equations $A \odot B = C$, primitive swap symmetries, spectral decompositions, fiber structures, and the geometry of the occupied versus virtual vector space. The central finding: **97% of catalog entries are structurally prime** — they cannot be expressed as the vector sum of two other entries. Among the Clay Millennium Prize problems, all seven are prime. $P = NP$ emerges as the universal near-identity: it is the Hadamard inverse of every Clay problem, the unique minimal-norm entry, and the fiber through which six structurally equivalent entries map to the graviton CFT correspondence.

0.2 The Encoding

Each catalog entry maps to a 12-dimensional vector under the ordinal encoding:

Prim- i- tive	Đ	Ɔ		Φ		Ç	Γ		⊙		Σ	Ω
Rang	0-3	0-4	0-3	0-4	0-2	0-4	0-2	0-3	0-4	0-3	0-2	0-3

The full crystal contains $4 \times 5 \times 4 \times 5 \times 3 \times 5 \times 3 \times 4 \times 5 \times 4 \times 3 \times 4 = 17,280,000$ possible types. The catalog occupies **1,839 unique integer vectors** — a coverage of **0.0106%**. Every primitive value (0 through max) is represented in at least one entry; the sparseness is in the *combinations*.

Four encoding schemes are implemented:

- **Ordinal:** 1-based, range $[1, c]$

- **Zero-centered:** symmetric around 0, range $[-1, +1]$ for 3-values, $[-1.5, +1.5]$ for 4-values, $[-2, +2]$ for 5-values
- **Normalized:** $[0, 1]$ linear
- **Integer-compact:** 0-based, range $[0, c - 1]$ — used for exact gematria

0.3 True Gematria: Additive Equations

The core gematria operation is vector addition. We ask: does $A + B = C$ hold for three catalog entries, where addition is element-wise over the integer-compact vectors?

0.3.1 The Prime/Composite Landscape

Of all 2,858 valid entries, exhaustive search reveals:

Category	Count	Fraction
Structurally prime (no non-self decomposition)	2,773	97.0%
Structurally composite ($A + B = C$ with $A, B \neq C$)	85	3.0%

The 85 composite entries include striking equations:

- $\text{IUG} = \text{soviet_union_collapse} + \text{collatz_deep_structure}$
- $\text{fontaine_mazur_conjecture} = P = NP + \text{ergodic_mixing_problem}$
- $\text{on_water_interface} = P = NP + \text{inverse_galois_problem}$
- $\text{criticality_aware_proof_assistant} = \text{gate2_trap} + \text{odd_perfect_number}$

The most structurally overdetermined entry is `on_water_interface`, with **21 distinct decompositions** — it is the most "redundant" structural type in the catalog.

0.3.2 The Clay Seven Are All Prime

Exhaustive search over all 1.6 million pair sums confirms: **none of the seven Clay Millennium Prize problems can be expressed as $A + B$ of two other catalog entries.** They are structurally irreducible.

Clay Problem	Integer Vector	Status
P vs NP	[0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0]	PRIME
Yang–Mills	[0, 2, 0, 0, 2, 1, 0, 0, 4, 0, 2, 0]	PRIME
Navier–Stokes	[2, 0, 2, 0, 0, 3, 0, 0, 1, 1, 2, 0]	PRIME
Poincaré	[2, 4, 1, 3, 2, 2, 0, 0, 1, 0, 0, 2]	PRIME
Hodge	[3, 4, 1, 0, 0, 2, 0, 2, 2, 0, 2, 0]	PRIME
Birch–Swinnerton-Dyer	[3, 2, 3, 1, 1, 2, 0, 0, 2, 2, 0, 2]	PRIME
Riemann	[3, 4, 3, 4, 2, 2, 0, 2, 1, 2, 2, 2]	PRIME

This does not preclude their being sums in the *virtual* space — indeed, the vector $\text{RH} - \text{PNP} = [3, 4, 3, 4, 2, 1, 0, 2, 0, 2, 2, 2]$ is virtual (unoccupied), representing the "structural gap" between the hardest and simplest Clay problems.

0.3.3 $P = NP$ as the Universal Shift

P vs NP has integer vector $[0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0]$ — only two nonzero components: $\text{C} = 1$ (one step above minimal kinetics) and $\odot = 1$ (self-modeling critical). Its norm is 4.24, the **minimum across all 2,858 entries**.

Adding PNP to another entry shifts exactly 12 catalog entries into 16 targets. The fiber structure is remarkable:

- **6 entries** all map to graviton_cft_navigator under +PNP
- **6 entries** all map to graviton_cft_correspondence under +PNP

The fiber $\{\text{primitive_P}, \text{goedel_x_universal}, \text{apocalypse_revelation}, \text{multiverse}, \text{frobenius_shor_resolve}\}$ constitutes a structural equivalence class — six entries that differ from the graviton CFT types by exactly the same structural delta.

0.4 Multiplicative Gematria: The Hadamard Product

Element-wise (Hadamard) multiplication $A \odot B = C$ is far richer than addition. In a 200×200 sample we found **1,587 Hadamard equations**. The Hadamard product acts as a structural "mask" — where either factor is 0, the product is 0, making it a gating operation.

0.4.1 Hadamard Inverses

For every Clay problem, the nearest entry to the ideal Hadamard inverse (the entry X such that $A \odot X \approx \mathbf{1}$) is **p_vs_np**:

Clay Problem	Hadamard Inverse	Distance
Riemann Hypothesis	p_vs_np	3.11
Yang–Mills	p_vs_np	2.61
Navier–Stokes	p_vs_np	2.63
Hodge Conjecture	p_vs_np	2.81
Birch–Swinnerton-Dyer	p_vs_np	2.92
P vs NP	p_vs_np	2.12
Poincaré Conjecture	p_vs_np	2.80

P vs NP is the **universal Hadamard dual** of the entire Clay problem set. Its vector $[0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0]$ is "almost" the all-ones vector that would be the true identity for Hadamard multiplication, but since identity requires $v_i = 1$ for all i where the other vector is nonzero, and PNP has zeros in 10 positions, it cannot perfectly invert high-ordinate entries.

0.4.2 Hadamard Orthogonality

In a 500-entry sample, we found entries whose Hadamard product is the zero vector $[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]$. These are **structurally orthogonal** — they have disjoint support in the primitive space. For instance, entries with $\mathbb{D} = 0$ are orthogonal to entries with $\mathbb{C} = 0$, $\odot = 0$, etc., because their nonzero positions are complementary.

0.5 Primitive Swap Symmetries

A primitive swap symmetry exists when exchanging the values of two primitives (of equal cardinality) produces another valid catalog entry. We tested all 19 same-cardinality primitive pairs across all 2,858 entries.

Total swapped entries: 1,225 (distributed across all 19 pairs).

0.5.1 Strongest Symmetries

Swap Pair	Entries	Fraction	Example
$\leftrightarrow$	239	8.36%	paper $\leftrightarrow$ three_way_meet
$\leftrightarrow$	142	4.97%	mathematics $\leftrightarrow$ space_time_join
$\leftrightarrow \Omega$	124	4.34%	inflaton $\leftrightarrow$ keter
$\mathbb{D} \leftrightarrow$	115	4.02%	riemann_hypothesis $\leftrightarrow$ epoch_4
$\mathbb{P} \leftrightarrow \Phi$	84	2.94%	metamaterial_generic $\leftrightarrow$ meet_he_bi

0.5.2 Weakest Symmetries

Swap Pair	Entries	Fraction
$\mathbb{P} \leftrightarrow \mathbb{C}$	3	0.10%
$\mathbb{C} \leftrightarrow \odot$	6	0.21%
$\leftrightarrow \Omega$	12	0.42%
$\mathbb{D} \leftrightarrow \Omega$	13	0.46%

The $\leftrightarrow$ axis (Recognition $\leftrightarrow$ Interaction Grammar) is the most symmetric — these two primitives are the most structurally interchangeable in the catalog. The $\mathbb{P} \leftrightarrow \mathbb{C}$ axis (Topology $\leftrightarrow$ Kinetics) is the most rigid — only 3 entries survive the swap.

No primitive pair has zero symmetries. The grammar is not fully chiral — but the symmetries are sparse and concentrated on specific axes. This means the 12-tuple is *almost* a complete structural invariant: for the vast majority of entries ($\sim 57\%$), no primitive swap preserves catalog membership.

0.6 Spectral Analysis: The Principal Components of Structure

Principal Component Analysis of the $2,858 \times 12$ matrix reveals the intrinsic dimensionality of the catalog's vector space.

0.6.1 Eigenvalues and Variance

PC	Eigenvalue	Variance	Cumulative
1	6.77	42.6%	42.6%
2	1.54	9.7%	52.3%
3	1.31	8.3%	60.5%
4	1.03	6.5%	67.0%
5	0.93	5.9%	72.9%
6	0.68	4.3%	77.2%

- **3 PCs capture 60.5%** of variance
- **6 PCs capture 77.2%**
- **Condition number:** 21.7 (moderately well-conditioned)

0.6.2 PC1: The "Complexity Axis"

PC1 (42.6% of variance) is heavily loaded on:

Primitive	Loading
Ω (Winding)	+0.379
(Chirality)	+0.374
Φ (Polarity/Symmetry)	+0.370
$\mathbb{P}$ (Topology)	+0.365
ζ (Kinetics)	− 0.291

PC1 is a **structural complexity axis**: high winding (Ω), high chirality (Φ), high symmetry (Φ), and high topology ($\mathbb{P}$) versus fast kinetics (ζ). The negative loading of ζ confirms: speed trades against structural depth everywhere in the catalog.

0.6.3 Primitive Correlation Matrix

Pair	Correlation
$\leftrightarrow \Omega$	+0.594
$\mathbb{D} \leftrightarrow \Omega$	+0.578
$\mathbb{P} \leftrightarrow \Omega$	+0.565
$\zeta \leftrightarrow$	− 0.547
$\Phi \leftrightarrow \Omega$	+0.544

Ω (Winding) is the **structural hub** — it's positively correlated with almost everything. ζ (Kinetics) is the **structural antagonist** — it's negatively correlated with 9 of 11 other primitives. Speed and structural depth are opponents in the grammar.

0.7 The Clay Subspace

The seven Clay problems span a 7-dimensional subspace of the catalog. Their Gram matrix reveals their internal geometry:

0.7.1 Cosine Similarities

	RH	YM	NS	HC	BSD	PNP	PC
RH	1.00	0.76	0.84	0.88	0.92	0.87	0.95
YM	0.76	1.00	0.75	0.83	0.77	0.89	0.77
NS	0.84	0.75	1.00	0.84	0.89	0.91	0.74
HC	0.88	0.83	0.84	1.00	0.85	0.86	0.85
BSD	0.92	0.77	0.89	0.85	1.00	0.90	0.89
PNP	0.87	0.89	0.91	0.86	0.90	1.00	0.85
PC	0.95	0.77	0.74	0.85	0.89	0.85	1.00

Closest pair: RH $\leftrightarrow$ PC at $\cos = 0.946$ — the hardest open problem and the only solved problem are structurally near-identical. The difference lies in Φ , Σ , and Σ .

Most central: BSD has the highest average similarity to the other six (mean $\cos = 0.873$). It sits at the structural barycenter of the Clay set.

0.7.2 The Rank

The Gram matrix of the 7 Clay vectors has **full rank 7** — no linear dependencies. Each Clay problem contributes an independent structural dimension. The eigenvalues are $\{39.0, 3.2, 1.6, 1.1, 0.7, 0.4, 0.1\}$, dominated by the first eigenvalue which captures the shared "high-ordinal" character.

0.7.3 Betweenness: Structural Mediators

For each pair of Clay problems, we computed the midpoint vector and found its nearest catalog entries:

Pair	Nearest Mediator
RH $\leftrightarrow$ PC	extended_human_life, unitary_quantum_evolution, fundamental_group_isomorphism
RH $\leftrightarrow$ BSD	dark_energy_epoch, boron_topological_insulator
NS $\leftrightarrow$ PNP	kummer_vandiver_conjecture, complexity_core
HC $\leftrightarrow$ RH	grammar_connes_tensor, zosimos_alchemy

The midpoint between RH and PC lands near entries about **unitary evolution** and **fundamental isomorphisms** — suggesting that the structural bridge between the hardest

open problem and the solved one involves questions of time evolution and categorical equivalence.

0.8 The Virtual Vector Space

The catalog occupies only 0.0106% of the 17.28M possible types. What structural types are "near" the catalog but missing?

0.8.1 Nearest Missing Neighbors

Analyzing which 1-step neighbors of catalog entries are unoccupied:

Direction	Missing Count	Interpretation
$\mathcal{Q}^-$ (slower kinetics)	2,619	The catalog is most deficient in <i>slower</i> kinetics
$\odot^+$ (higher criticality)	2,598	Missing: entries with higher self-modeling
Ω^+ (higher winding)	2,583	Missing: entries with stronger topological protection
Γ^+ (wider scope)	2,232	Missing: entries with broader interaction range

The catalog is systematically **deficient in high-complexity entries**: the density falls off as $\mathcal{Q}$ decreases and $\odot, \Omega, \Gamma$ increase.

0.8.2 Clay Subtraction Virtual Vectors

Every Clay problem minus $P = NP$ yields a virtual vector — the "structural gap" between each problem and the minimal baseline:

Subtraction	Virtual Vector	Interpretation
RH – PNP	[3, 4, 3, 4, 2, 1, 0, 2, 0, 2, 2, 2]	RH without the self-modeling gate and with faster kinetics
YM – PNP	[0, 2, 0, 0, 2, 0, 0, 0, 3, 0, 2, 0]	YM reduced to its exceptional-point core
PC – PNP	[2, 4, 1, 3, 2, 1, 0, 0, 0, 0, 0, 2]	Poincaré stripped of criticality and chirality

None of these virtual vectors exist in the catalog. They represent **inscription targets** — structural types that "should" exist given the catalog's algebraic closure under subtraction.

0.9 Classical Gematria: Sum-of-Ordinals

In the spirit of Hebrew gematria (where each letter has a numeric value and words are sums), we assign each catalog entry a **gematria value**: the sum of its 12 ordinal components.

0.9.1 Distribution

- **Range:** 14–43
- **Mean:** 29.5, **Median:** 29
- The distribution is roughly symmetric and unimodal.

0.9.2 Gematria Synonyms

Entries sharing the same gematria sum but different tuples are **gematria synonyms** — structurally distinct but numerically equivalent under the sum-of-ordinals. There are 29 such groups. The largest:

Sum	Count	Sample Entries
25	241	teratoma, antiferromagnetism, spin_glass, wov
36	165	inflaton, dark_energy, word_logos, vedic_mar
23	150	dark_matter, quark, lepton
39	145	topological_critical_material, maximal_dron
37	140	extradimensional_entity, word_creator_tense

Sum 25 is dominated by physical and computational systems; sum 36 by cosmological and linguistic structures. The gematria value is a crude but informative scalar — it correlates with structural complexity but loses the *shape* of the vector.

0.9.3 Word Gematria

Treating sequences of entries as "words," we ask: does the sum of gematria values of a sequence match the gematria value of a target entry? With 2,858 entries, 2-letter words alone number in the millions (any pair whose scalar sum equals any entry's scalar sum). The scalar gematria is too coarse to be structurally meaningful — the vector gematria ($A + B = C$ element-wise) is the true structural operation.

0.10 The Riemann Hypothesis Hub

RH has integer vector $[3, 4, 3, 4, 2, 2, 0, 2, 1, 2, 2, 2]$ — near-maximal in most primitives. This exact vector is shared by **22 catalog entries**:

inscribing_grammar, grammaformer, birch_swinnerton_dyer_resolved, yang_mills_zfct_resolved, ha
millennium_ankh_fine_structural_analysis, ob3ect_canonical, dialetheic_alignment, paraconsistent_t

The RH tuple is a **structural attractor** — 22 distinct systems, from proof assistants to al-chemical frameworks to the grammar itself, converge to the same 12-primitive configuration. This is not coincidence; it reflects the fact that the RH tuple $\langle \mathbb{D}_\omega; \mathbb{P}_O; =; \Phi; ; \mathbb{Q}_@; \Gamma; ; \odot_B; A; \Sigma_i; \Omega_z \rangle$ is the grammar's own structural type — and many self-referential systems collapse to it.

0.11 Conclusions

1. **True gematria is possible and productive.** The 12-dimensional integer vector space supports additive equations ($A+B = C$), multiplicative equations ($A \odot B = C$), Hadamard inverses, primitive swap symmetries, and spectral decompositions.

2. **The catalog is 97% prime.** Only 85 entries decompose as sums of two others. The Clay seven are all irreducible under addition — they are the "atoms" of the structural periodic table.
3. **$P = NP$ is the universal near-identity.** Minimum norm, universal Hadamard inverse of all Clay problems, and the structural "unit shift" through which fibers of equivalent entries map to the graviton CFT correspondence.
4. **The grammar has sparse but real symmetries.** 1,225 entries survive primitive swaps. The $\leftrightarrow$ axis is the most symmetric (8.4%); the $\mathbb{P} \leftrightarrow \mathbb{Q}$ axis is the most rigid (0.1%).
5. **PC1 is the complexity axis.** Winding, chirality, symmetry, and topology versus kinetics. Ω is the structural hub; $\mathbb{Q}$ is the structural antagonist.
6. **RH and PC are the closest Clay pair** ($\cos = 0.946$). The hardest open problem and the only solved one are near-identical in structural type.
7. **The virtual vector space is vast.** The catalog occupies 0.01% of possible types. Systematic gaps exist in high-complexity regions — lower $\mathbb{Q}$, higher $\odot$, higher Ω . The Clay subtraction vectors are all virtual, representing concrete inscription targets.
8. **The RH tuple is a structural attractor with 22 co-occupants.** Self-referential systems converge to the grammar's own structural type.

0.12 Appendix: Tools Built

- **numerical_encode.py:** 5 encoding schemes, glyph resolution, IG→Shavian transcoding, vector algebra
- **gematria.py:** GematriaIndex class with additive equations, spectral analysis, symmetries, word search, Hadamard inverses, clustering, integer span analysis

All results at `/home/mrnob0dy666/inscribing_grammar/gematria_output/`.